Vaishnavi Pulavarthi

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OBJECTIVE

Electrical Engineer, specializing in **Embedded Systems**, **Robotics**, and **Hardware Design**, ready to contribute to the development of autonomous systems and intricate hardware design and verification. I seek opportunities to further enhance my skills and actively pursue breakthroughs in my field.

EDUCATION

Master of Science in Electrical and Computer Engineering, August 2022 – May 2024 | Chicago, United States
University of Illinois at Chicago, GPA 4.0/4.0

Bachelor of Engineering in Electronics and Communication Engineering, July 2015 – May 2019 | Hyderabad, India
Osmania University, GPA 8.8/10.0

SKILLS

Python, C, Shell Scripting, Verilog HDL, Arduino, MATLAB, Altium PCB Design, Cadence Virtuoso, HTML3, SQL.

EXPERIENCE

Graduate Research Assistant, *University of Illinois at Chicago* December 2022 – present | Chicago, United States

• Utilizing prompt engineering techniques to enhance LLM's proficiency in interpreting Verilog HDL and assertions, focusing on effective training methods and engaged in in-depth source code analysis using Goldmine and employed it for training the LLM (**LLAMA2**).

Systems Operations Analyst, *Wells Fargo India Pvt. Ltd* July 2020 – July 2022 | Hyderabad, India

• Handled **Black Duck** and IBM **Urban Code Deploy,** including infrastructure setup for application servers, and automated operational tasks on IBM Urban Code Deploy using REST API, resulting in a **5%** increase in operational efficiency.

Apps Programmer/Analyst, *Wells Fargo India Pvt. Ltd.* July 2019 – July 2020 | Hyderabad, India

• Leveraged tools such as **Jenkins**, **UCD**, and **Black Duck** to implement **CI/CD** and **DevOps**, achieving **25%** faster project deployments, while also efficiently coordinating Black Duck operations for all Wells Fargo projects.

LABORATORY WORK

Robotics and Algorithms, *University of Illinois at Chicago* March 2023 – May 2023

• Programmed an autonomous car to detect obstacles, follow **ArUco** markers using a camera, implement odometry algorithms on **Raspberry Pi**, and integrate these components to enable bug algorithms for efficient navigation.

Mechatronic Embedded Systems, *University of Illinois at Chicago* January 2023 – April 2023

• Designed and implemented an autonomous car that integrates motors, servos, line camera, motor controller and boost converter on a PCB utilizing an **FRDM** micro-controller board, algorithms for line detection and autonomous navigation.

PROJECTS

Image Analysis and Computer Vision, *University of Illinois at Chicago* February/2023 – April/2023

• Programmed a two - dimensional equiripple **FIR** filter in **MATLAB** and implemented on a dataset to analyze the **PSNR** values of original, noise and refined images.

• Evaluated **YOLOv7** and **v8** versions of the object detection algorithm on a diverse dataset and analyzed detection accuracy, speed, efficiency while exploring potential applications.

Indoor Positioning System using Bluetooth, *Osmania University* December 2018

• Engineered circuits for signal propagation and reception, improving signal strength accuracy within the grid, while analyzing Bluetooth device data and developing sophisticated algorithms that boosted positioning accuracy by **7%**.

• Developed and deployed a fully functional application to precisely locate individuals inside a building using Bluetooth technology, enhancing indoor navigation by **8%**.

Notice board display using Wi-Fi module, *Osmania University and C-DAC* May 2018

• Collaborated as an intern at **C-DAC** to create a project where the microcontroller effectively received signals from a Wi-Fi module, leading to **12%** faster display response time and improved data accuracy.

• Mastered Embedded systems, ARM processors, and various peripherals, including ADC, Watchdog timer, and Systick Timer, through hands-on programming during a 4-week C-DAC internship.

ACHIEVEMENTS AND ACTIVITES

Event Coordinator, *University of Illinois at Chicago* March 2022

• Coordinated with Engineering Career Center staff and student volunteers to optimize event registration, achieving a **30%** reduction in processing time, **25**% higher participant satisfaction.

Marketing Lead, *Quizzotic, OU* January 2018 - March 2018

• Directed the marketing team to create and execute strategies that attracted over 300 participants to Quizzotic and achieved 25% increase in event participation compared to the previous year through effective marketing and promotion effort.